

Validity Analysis: MMLU

Target context: North India Competitive Exam Aspirants (Hindi-Medium)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	1	Serious concern	high
Input Content 	1	Serious concern	high
Input Form 	1	Serious concern	high
Output Ontology 	1	Serious concern	high
Output Content 	1	Serious concern	high
Output Form 	2	Significant gaps	high
Average	1.2		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is a near-total mismatch for the North India Hindi-medium competitive exam deployment across five of six validity dimensions. Subject coverage is anchored in US/Western academic curriculum with zero Indian Polity, Indian History, Indian

Geography, Current Affairs, or Hindi-language content (the high-priority domains identified in elicitation). Individual items are culturally embedded in US institutional, legal, demographic, and moral contexts; where India does appear, it is filtered through colonial or Western-pollster framing incompatible with the NCERT/Bipin Chandra-aligned exam-prep tradition. The benchmark is entirely English with no Devanagari content, conflicting with the deployment's Hindi-dominant input requirement. Output is label-only with no rationale evaluation, while the deployment requires substantive Hindi explanatory feedback. Annotator pool is undocumented but US-anchored; moral_scenarios labels are explicitly framed in US 2020 norms. Output form is the only dimension with a partial structural match (MCQ format), but even that is undermined by calibration failures and absence of generative-quality measurement. Indian-grounded alternatives (MILU, ParamBench, IndicEval, IndicMMLU-Pro) exist and should be primary; MMLU at best provides weak proxy signal on STEM/reasoning subdomains.

Practical Guidance

What This Benchmark Measures

MMLU measures broad English-language multitask classification accuracy on a US/Western academic curriculum, with strongest signal on culturally-neutral STEM (mathematics, physics, computer science) and weakest on humanities, law, and morality. For the NI-CEA deployment, MMLU can provide only weak proxy evidence for the Mathematics & Reasoning domain (one of several high-priority areas) and provides no signal at all on Indian Polity, Indian History, Indian Geography, Current Affairs, Hindi proficiency, or Hindi explanatory feedback quality.

Construct Depth

MMLU probes breadth across 57 subjects but at modest depth (5-shot prompts, label-only scoring [Q34, Q44]). For the deployment's core constructs — India-grounded factual knowledge, Hindi-medium comprehension, and Hindi pedagogical explanation — MMLU's depth is effectively zero. The English-vs-Hindi performance gap documented in IndicEval [WEB-16] confirms that MMLU's English-medium scores do not predict Hindi-medium ability even on overlapping content.

What Else You Need

Substantial supplementation is required. (1) For Input Ontology / Input Content, use MILU [WEB-13, WEB-14] and ParamBench [WEB-15] as primary benchmarks; they cover UPSC/SSC-relevant Indian curriculum content in Hindi. (2) For Input Form, use IndicEval [WEB-16] or IndicMMLU-Pro [WEB-18] to measure Hindi-medium degradation. (3) For Output Ontology / Output Form (Hindi explanatory feedback), commission a custom rubric — no existing benchmark covers this gap [WEB-21]. (4) For Output Content, conduct Indian-SME annotation review on any MMLU items used. (5) For Current Affairs, build a continuously-updated PIB/Yojana-sourced evaluation set, since static benchmarks cannot cover this domain.

ID	Evidence
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-14	https://huggingface.co/datasets/ai4bharat/MILU
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)
WEB-18	https://arxiv.org/abs/2501.15747
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's 57-subject taxonomy is anchored in a US/Western academic curriculum [\[Q1, Q8, Q16\]](#) with subjects calibrated to US bar-exam, USMLE, and US-foreign-policy concerns [\[Q17, Q49\]](#). None of the high-priority Indian competitive exam domains identified in the elicitation (Indian Polity & Constitution, Indian History, Current Affairs, Indian Geography, Hindi language proficiency) is represented. Dataset analysis confirms across ~200 sampled items zero coverage of Indian Polity, Indian History, or Indian Geography; the only South Asian content is two Jainism items in world_religions and one colonial-framed 1857 item. Partial overlap exists for STEM/mathematics and reasoning, which are universally applicable, but this is a small fraction of the deployment's high-priority surface.

ID	Evidence
Q1	"We propose a new test to measure a text model's multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more." (p.1)
Q8	"The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability." (p.1)
Q16	"We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B." (p.3)
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q49	"Figure 6 shows the accuracy of GPT-3 (few-shot) and UnifiedQA for all 57 tasks. It shows the both models are below expert-level performance for all tasks, with GPT-3's accuracy ranging from 69% for US Foreign Policy to 26% for College Chemistry. UnifiedQA does best on marketing, with an accuracy of 82.5%." (p.6)

Strengths

- Mathematics, abstract algebra, computer science, and conceptual physics items are culturally neutral and provide partial proxy signal for SSC quantitative aptitude / UPSC CSAT reasoning [\[DATASET-D27, D29, D30, D40\]](#).
- NCERT-overlapping science content (biology, virology, evolution) gives weak-but-positive signal for Indian GK science sections [\[DATASET-D41, D42\]](#).

ID	Evidence
D27	Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context
D29	Let $G = (V, E)$ be a connected, undirected graph ... P_1 can be solved in polynomial time but P_2 is not known to be solvable in polynomial time. — Algorithm complexity theory — culturally neutral STEM content
D30	Find the characteristic of the ring $\mathbb{Z}_3 \times 3\mathbb{Z}$. — Abstract mathematics — entirely culture-independent
D40	A wave transfers ... energy — Basic physics principle — universally applicable
D41	The sequence of amino acids in hemoglobin molecules of humans is more similar to the hemoglobin of chimpanzees than it is to the hemoglobin of dogs. — Biology/evolution — curriculum-neutral STEM content

D42	Viruses have encouraged us to change our world, as we have now: Eradicated smallpox — General science fact — applicable to Indian science GK
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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per elicitation: General Knowledge (India-centric), Current Affairs, Indian History (ancient/medieval/modern), Indian Polity & Constitution, Mathematics & Reasoning, Hindi language proficiency, with supplementary Indian Geography and Indian Economy.

IO-2: MMLU omits all India-centric categories. The history component claims wide geographic scope [Q27] but sampled items show European and US history exclusively [DATASET-D4, D5, D12, D13, D14]. No Hindi proficiency, no Indian Polity, no Current Affairs subject [Q104].

IO-3: MMLU includes US-jurisdiction-specific categories (professional_law calibrated to US bar exam [Q17], professional_accounting on US CPA standards [DATASET-D32], us_foreign_policy [DATASET-D7, D8, D9], moral_scenarios anchored to US 2020 norms [DATASET-D18, D19]) that are construct-irrelevant for the deployment.

IO-4: Content validity is severely harmed: the benchmark's subject taxonomy [Q104] does not map onto the UPSC/SSC syllabus published at upsc.gov.in [WEB-5] or ssc.gov.in [WEB-6]. Indian-curriculum-grounded benchmarks (MILU, ParamBench, IndicEval) demonstrate that this gap is addressable but absent from MMLU itself.

ID	Evidence
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q27	"Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects." (p.4)
Q104	"Table 2: Summary of all 57 tasks." (p.15)
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 ... The language of 'protest' that Calhoun used ... was similar to the language of which of the following political positions? — US history question with primary source passage — entirely irrelevant to Indian History
D5	Between 1820 and 1854, the greatest number of immigrants to the United States came from ... Ireland — US immigration history — no relevance to Indian competitive exams
D7	Why might the 'Philadelphian System' be linked to the idea of American exceptionalism? — US foreign policy / exceptionalism — irrelevant to Indian deployment
D8	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — no analogue in UPSC/SSC syllabus
D9	In American government, the power to declare war rests with ... Congress — US government structure question — not Indian Polity
D12	The Scribbling-Machines have thrown thousands of your petitioners out of employ ... Leeds Woolen Workers Petition, 1786 — British Industrial Revolution primary source — European history with no South Asian content
D13	What is tolerance? ... Of all religions, the Christian ought doubtless to inspire the most tolerance ... Voltaire, Letters on the English Nation, 1733 — Enlightenment Western philosophy — Western canon, irrelevant to Indian History syllabus
D14	The substitution of Plato for the scholastic Aristotle ... Cosimo and Lorenzo de Medici ... the Florentine Academy ... Neoplatonism — Italian Renaissance intellectual history — no Indian History coverage

D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams
WEB-5	UPSC official syllabus published at upsc.gov.in (https://upsc.gov.in/examinations/active-examination)
WEB-6	SSC CGL/CHSL syllabus at ssc.gov.in (https://ssc.gov.in/)
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Information Gaps

- Whether any subset of MMLU's miscellaneous or global_facts items can be systematically mapped to UPSC GS general-awareness sub-topics.

Requires Expert Verification

- Mapping of MMLU mathematics items to specific UPSC CSAT / SSC CGL quantitative aptitude topics by an Indian exam-prep subject-matter expert.

Remediation

Gap 1: Zero coverage of Indian Polity, Indian History, Indian Geography, Current Affairs, Hindi proficiency — the core UPSC/SSC syllabus.

Recommendation: Replace MMLU as primary evaluation with MILU [WEB-13] and ParamBench [WEB-15], which cover UPSC/UGC-NET-relevant Indian content. Use MMLU at most as a STEM-subset proxy after filtering to abstract_algebra, college/high_school_mathematics, conceptual_physics, college_computer_science.

ID	Evidence
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Individual MMLU items are culturally anchored in US/Western contexts: US Congress and constitutional history [\[DATASET-D1, D2, D3\]](#), US bar-exam tort scenarios [\[DATASET-D10, D11\]](#), Western philosophical canon [\[DATASET-D16, D17\]](#), US Social Security and birth-control demographics [\[DATASET-D20, D21\]](#), US measurement units in mathematics [\[DATASET-D25, D28\]](#), and CPA/MCAT institutional referents [\[DATASET-D32, D48\]](#). The benchmark assumes models acquired requisite knowledge from 'vast quantities of diverse Internet text' [\[Q60\]](#), which is overwhelmingly English-Western. The deployment requires both pan-India and North India-specific factual content (Chhath Puja, Panchayati Raj, tehsils, NCERT-aligned narratives) per elicitation Q2, none of which appears in the sampled items. Where India is referenced, it is via Pew Research statistics [\[DATASET-D23, D24\]](#) or colonial-British framing of the 1857 uprising [\[DATASET-D15\]](#) — both incompatible with NCERT/Bipin Chandra-aligned exam preparation.

ID	Evidence
Q2	"To attain high accuracy on this test, models must possess extensive world knowledge and problem solving ability." (p.1)
Q60	"A major distinction between our benchmark and previous multitask NLP benchmarks is that we do not require large training sets. Instead, we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet. For this reason we assess pretrained models in a zero-shot, few-shot, or transfer setting and we provide a dev, val, and test set for each task." (p.7)
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law question — irrelevant to Indian Polity
D2	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US Congress-specific legislation — not Indian constitutional content
D3	What power was granted to the states by the Articles of Confederation but not by the Constitution? — US Articles of Confederation — no Indian Polity equivalent
D10	A truck driver was assigned to drive a huge tractor-trailer loaded with logs ... the girl's parents ... assert a claim against the lumber company — US tort law scenario — calibrated against US bar exam, not Indian law
D11	An avid baseball fan ... forwarded the club a \$50 registration deposit ... In a restitutionary action, can the executor of the fan's estate ... recover ... the \$2,500 paid to the club? — US contract law with baseball cultural context — no Indian law equivalent
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D16	According to Parfit, the obligation to give priority to the welfare of one's children is: agent-relative. — Western analytic philosophy (Parfit) — not covered in UPSC/SSC philosophy or ethics sections
D17	Anscombe claims that on Sidgwick's view, the badness of an action must be estimated in light of: its expected consequences. — Western analytic moral philosophy (Sidgwick, Anscombe) — entirely Western canonical tradition
D20	The age at which one can receive full benefits from social security ... Has been increasing — US Social Security system — irrelevant to Indian context
D21	Which of the following is the most common method of birth control for married couples in the U.S. — Explicitly US-specific demographic question
D23	As of 2017, what fraction of the population in India used the internet in the past three months? ... 26% — One of very few India-referencing questions — but tests dated Western-sourced statistical trivia, not Indian GK curriculum

D24	As of 2018, about what percentage of Indians say that having an increasing number of people of many different races ... in their country makes it a better place to live ... 68% — India-referencing global survey question — but sourced from Pew/Western pollsters, not Indian exam curriculum
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams
D48	When preparing for the MCAT exam, a student begins studying electrochemical cells. — References MCAT (US Medical College Admission Test) — US-specific academic context

Strengths

- Pure-mathematics items (abstract algebra, probability) are content-culturally neutral and transfer cleanly [DATASET-D27, D30].
- Some basic NCERT-overlapping science facts (smallpox eradication, evolution) are factually correct in an Indian curriculum context [DATASET-D41, D42].

ID	Evidence
D27	Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context
D30	Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$. — Abstract mathematics — entirely culture-independent
D41	The sequence of amino acids in hemoglobin molecules of humans is more similar to the hemoglobin of chimpanzees than it is to the hemoglobin of dogs. — Biology/evolution — curriculum-neutral STEM content
D42	Viruses have encouraged us to change our world, as we have now: Eradicated smallpox — General science fact — applicable to Indian science GK

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires India-centric cultural and administrative knowledge (festivals like Chhath Puja, tehsils, mandals, North Indian historical figures, NCERT-aligned narratives) per elicitation Q2; MMLU items do not address these.

IC-2: No — culturally sensitive content in MMLU (moral_scenarios explicitly framed as 'US 2020 moral standards' [DATASET-D18, D19]; jurisprudence using US product liability [DATASET-D33]) does not align with Indian deployment culture.

IC-3: Yes — extensive Western-specific knowledge requirements: US tax law [Q152], US bar-exam scenarios [Q153], US CPA standards [DATASET-D32], MCAT framing [DATASET-D48], US traffic conventions [DATASET-D45], and US customary measurement units in arithmetic [DATASET-D25, D28].

IC-4: INSUFFICIENT DOCUMENTATION — the paper does not document any review of items by Indian regional annotators; would need a systematic Indian SME audit to identify all culturally sensitive instances.

IC-5: Content validity is severely harmed: when India does appear [D15, D47], the framing privileges colonial historiography over the nationalist NCERT/Bipin Chandra narrative central to UPSC preparation, creating direct misalignment for exam-prep use.

ID	Evidence
Q2	"To attain high accuracy on this test, models must possess extensive world knowledge and problem solving ability." (p.1)
Q60	"A major distinction between our benchmark and previous multitask NLP benchmarks is that we do not require large training sets. Instead, we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet. For this reason we assess pretrained models in a zero-shot, few-shot, or transfer setting and we provide a dev, val, and test set for each task." (p.7)
Q152	"Krete is an unmarried taxpayer with income exclusively from wages. By December 31, year 1, Krete's employer has withheld \$16,000 in federal income taxes and Krete has made no estimated tax payments. On April 15, year 2, Krete timely filed for an extension request to file her individual tax return, and paid \$300 of additional taxes. Krete's year 1 tax liability was \$16,500 when she timely filed her return on April 30, year 2, and paid the remaining tax liability balance. What amount would be subject to the penalty for underpayment of estimated taxes? (A) \$0 (B) \$500 (C) \$1,650 (D) \$16,500" (p.24)
Q153	"The night before his bar examination, the examinee's next-door neighbor was having a party. The music from the neighbor's home was so loud that the examinee couldn't fall asleep. The examinee called the neighbor and asked her to please keep the noise down. The neighbor then abruptly hung up. Angered, the examinee went into his closet and got a gun. He went outside and fired a bullet through the neighbor's living room window. Not intending to shoot anyone, the examinee fired his gun at such an angle that the bullet would hit the ceiling. He merely wanted to cause some damage to the neighbor's home to relieve his angry rage. The bullet, however, ricocheted off the ceiling and struck a partygoer in the back, killing him. The jurisdiction makes it a misdemeanor to discharge a firearm in public. The examinee will most likely be found guilty for which of the following crimes in connection to the death of the partygoer? (A) Murder. (B) Involuntary manslaughter. (C) Voluntary manslaughter. (D) Discharge of a firearm in public." (p.24)
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams
D33	Bill purchased a can of Sipep from the Ajax Minimart ... In a strict product liability tort action against Ajax, Bill must prove ... — US-style product liability tort — uses American legal framework
D45	A flashing red traffic light signifies that a driver should do what? ... stop — US traffic law convention — minor but US-specific
D47	the deed was done by a subject race — by black men who dared to shed the blood of their masters ... British journalist William Howard Russell — 1857 mutiny through British colonial journalist's lens — if Indian content appears at all in MMLU, it is through a colonial/Western perspective
D48	When preparing for the MCAT exam, a student begins studying electrochemical cells. — References MCAT (US Medical College Admission Test) — US-specific academic context
WEB-20	Translated MMLU variants suffer cultural mismatch because content (US History, Law) remains Western-specific even in Hindi (https://arxiv.org/abs/2509.11570)

Information Gaps

- Full enumeration of India-relevant items beyond the ~200 sampled would require complete corpus review of all 14,079 test items.
- Annotator-side cultural assumptions are not documented in the paper.

Requires Expert Verification

- Indian SME review of moral_scenarios and jurisprudence items to determine extent of label flips under Indian normative frameworks.
 - Audit of how 1857 / colonial history items would be scored against NCERT-aligned answer expectations.
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Remediation

Gap 1: US measurement units (miles, feet) in mathematics; US institutional referents (CPA, MCAT, Social Security) introduce construct-irrelevant variance.

Recommendation: If MMLU mathematics items are reused, locally adapt unit conventions to SI/metric and replace US-institution-specific framings; alternatively, source quantitative-aptitude items directly from SSC CGL / UPSC CSAT past papers.

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU is entirely English-language, text-only, with a standardised English preamble for prompts [Q44] and English-only normalised input format for UnifiedQA [Q91]. Dataset metadata confirms 'languages: [en]' and monolingual content; no Devanagari script or Hindi vocabulary appears in any of the ~200 sampled items. The deployment requires Hindi-dominant interaction in Devanagari script with at most ~10% English code-mixing per elicitation Q4. The MCQ structural format (4 options A/B/C/D) is a partial structural match to UPSC/SSC exam formats, but this is overridden by total language and script mismatch. IndicEval [WEB-16] empirically demonstrates a large Hindi-vs-English performance gap (LLaMA 39.53% on UPSC Hindi Zero-Shot vs. ~84.88% Gemini on UPSC English CoT), confirming that English-medium MMLU scores cannot predict Hindi-medium performance.

ID	Evidence
Q4	"However, on every one of the 57 tasks, the best models still need substantial improvements before they can reach expert-level accuracy." (p.1)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q91	"UnifiedQA's input format is of the form QUESTION1 \n (A) CHOICE1 (B) CHOICE2 (C) CHOICE3 (D) CHOICE4</s> where questions and choices are normalized and made lowercase." (p.12)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)

Strengths

- Four-option MCQ structure (A/B/C/D) [Q44, Q83] is a structural match to UPSC Prelims, SSC CGL, and banking exam formats [DATASET strength 1].
- Text-only modality is appropriate for the deployment's text-based MCQ context (per elicitation deployment description, mobile/enterprise application is text-driven).

ID	Evidence
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Severe mismatch — MMLU input distribution is 100% English text [Q44, Q91]; deployment input distribution is Hindi (Devanagari) with up to ~10% English code-mixing.

IF-2: Regional infrastructure does support Devanagari rendering (Gboard, SwiftKey Hindi available) but the benchmark provides no Devanagari content to evaluate; Hindi tokenization quality is a documented live deployment risk [WEB-1].

IF-3: Form-relevant differences: MMLU uses no Devanagari script, no Hindi morphology, no code-mixed Hinglish; mathematics items use US customary units (miles, feet) rather than SI units used in Indian exams [DATASET-D25, D28].

IF-4: External validity is severely harmed: MMLU accuracy on English inputs provides no evidence about model performance on Hindi-medium inputs; IndicEval [WEB-16] documents catastrophic multilingual degradation.

ID	Evidence
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q59	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
Q91	"UnifiedQA's input format is of the form QUESTION1 \n (A) CHOICE1 (B) CHOICE2 (C) CHOICE3 (D) CHOICE4</s> where questions and choices are normalized and made lowercase." (p.12)
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention
WEB-1	Devanagari tokenization quality is a live deployment risk; even models trained on 20B Hindi tokens hallucinate or mix in English (https://arxiv.org/abs/2410.14815)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)

Information Gaps

- Quality of community Hindi MMLU translations (OpenAI MMMLU, IndicMMLU-Pro) is documented in web sources but not directly assessed here.

Requires Expert Verification

- Hindi linguistic review of any translated MMLU variant for register appropriateness (Manak Hindi vs. literal translation) for exam-prep context.

Remediation

Gap 1: 100% English input with no Devanagari script; deployment requires Hindi-dominant input.

Recommendation: Adopt IndicEval [WEB-16] for direct UPSC-Hindi performance measurement and IndicMMLU-Pro [WEB-18] for broader Hindi-translation assessment. Treat any English-MMLU score as upper-bound only; expect substantial Hindi-medium degradation.

ID	Evidence
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)
WEB-18	https://arxiv.org/abs/2501.15747

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's output ontology is a four-class A/B/C/D classification [Q83, Q44] with classification accuracy as the sole performance criterion [Q34, Q23]. Dataset analysis confirms a single ClassLabel field (0–3) with no rationale field. The deployment requires a fundamentally richer output taxonomy: a verdict (sahi/galat) plus a substantive Hindi-language explanation in standard Manak Hindi [elicitation Q3]. MMLU's design provides zero signal on the model's capacity to produce explanatory rationale, let alone in Hindi. The benchmark's stated rationale for the simple format ('a simple-to-evaluate test that measures classification accuracy' [Q23]) is internally coherent but structurally incompatible with the deployment's pedagogical-feedback requirement.

ID	Evidence
Q3	"We find that while most recent models have near random-chance accuracy, the very largest GPT-3 model improves over random chance by almost 20 percentage points on average." (p.1)
Q23	"Consequently, we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)

Strengths

- Verdict component (correct vs. incorrect) is a partial match — the A/B/C/D classification can be reduced to a binary correct/incorrect signal that does align with one component of the deployment output (the sahi/galat label) [Q83, DATASET concern 3].

ID	Evidence
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values

OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels (A/B/C/D) [Q83] partially match the deployment's required correctness verdict but entirely omit the explanation component. Deployment requires Hindi explanatory feedback per elicitation Q3.

OO-2: Missing categories: free-form Hindi explanation, distractor analysis (why each wrong option is wrong), and reference to canonical Indian sources (NCERT, Laxmikant). None are represented in MMLU's output schema.

OO-3: Output categories themselves are content-neutral (letter labels), but the underlying scoring rule treats US-anchored answer keys (e.g., moral_scenarios under US 2020 norms [DATASET-D18, D19]) as correctness ground truth, encoding non-regional values.

OO-4: A stakeholder-driven output taxonomy redesign is required: minimum {verdict_label, hindi_explanation_text, optional_distractor_analysis} with a separate rubric for explanation quality. No existing benchmark covers this — the rubric gap is documented as unresolved [WEB-21].

OO-5: Structural and content validity are severely harmed: MMLU's output structure measures only one of two required output components and applies decision rules anchored in US norms.

ID	Evidence
Q3	"We find that while most recent models have near random-chance accuracy, the very largest GPT-3 model improves over random chance by almost 20 percentage points on average." (p.1)
Q23	"Consequently, we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q62	"Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining." (p.8)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Information Gaps

- No published rubric for Hindi exam-prep explanation quality [WEB-21] — cannot quantify the magnitude of the output-ontology gap.

Requires Expert Verification

- Indian exam-prep pedagogues need to define an explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) for the deployment.

Remediation

Gap 1: Label-only A/B/C/D output cannot evaluate the required Hindi explanatory feedback.

Recommendation: Commission a Hindi exam-prep explanation-quality rubric (factual accuracy, distractor analysis, Manak Hindi register, NCERT alignment) with Indian SME panel — this is a documented gap in current literature [WEB-21] and must be built bespoke for the deployment.

ID	Evidence
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The paper is silent on annotator demographics, qualifications, IAA protocols, and label-validation procedures — Output Content concerns are explicitly NOT DOCUMENTED in the benchmark write-up. Questions were assembled by US graduate/undergraduate students [\[Q17\]](#) and validated against US-standard answer keys. No Indian SMEs, UPSC/SSC specialists, or Hindi-medium educators participated. For India-relevant items that do appear ([DATASET-D15](#) 1857 uprising), the answer keys privilege colonial historiography over NCERT/Bipin Chandra nationalist framing, directly conflicting with what Indian exam stakeholders would judge correct. moral_scenarios labels are explicitly anchored to 'ordinary moral standards in the US as of 2020' [\[DATASET-D18, D19\]](#), producing systematic OC misalignment for the target population. Even the strongest Indian-grounded benchmarks (ParamBench, MILU) only partially document Indian-SME involvement [\[WEB-15, WEB-13\]](#); MMLU documents none.

ID	Evidence
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Strengths

- For purely-cultural-neutral STEM items (abstract algebra, conceptual physics), label correctness is unlikely to depend on annotator culture — these labels probably transfer to Indian stakeholders [\[DATASET-D27, D30, D40\]](#).
- Memorization analysis suggests low risk that exact questions were leaked into pretraining [\[Q98, Q99\]](#), so labels are not artifacts of test contamination.

ID	Evidence
Q98	"However, in Figure 13, we see that accuracy and question entropy are not positively correlated, suggesting that the test's low-entropy questions do not correspond to memorized (and thereby correctly predicted) answers." (p.14)
Q99	"This suggests that our exact questions were not memorized." (p.14)
D27	Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context
D30	Find the characteristic of the ring $\mathbb{Z}_3 \times 3\mathbb{Z}$. — Abstract mathematics — entirely culture-independent

D40	A wave transfers ... energy — Basic physics principle — universally applicable
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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect US-educated annotator perspectives [Q17]; for India-centric content, Indian stakeholder perspectives are unrepresented.

OC-2: Significant disagreement is plausible for moral_scenarios (US 2020 norms vs. Indian norms) [DATASET-D18, D19], for high_school_world_history items framed via colonial sources [DATASET-D15, D47], and for any Indian-content item validated against non-Indian answer keys.

OC-3: INSUFFICIENT DOCUMENTATION — paper does not include a Datasheet or Data Statement section addressing annotator demographics; only [Q17] notes 'graduate and undergraduate students' without further detail.

OC-4: Re-annotation by representative Indian annotator pool is necessary for any India-deployment use; ParamBench [WEB-15] provides a partial template but does not cover MMLU items.

OC-5: INSUFFICIENT DOCUMENTATION — aggregation methods for label disagreements are not described; cannot assess minority-perspective erasure.

OC-6: Convergent and external validity are severely harmed: MMLU labels do not correlate with Indian-stakeholder judgments on culturally embedded items, and US-norm-anchored labels do not generalize to Indian deployment context.

ID	Evidence
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
D47	the deed was done by a subject race — by black men who dared to shed the blood of their masters ... British journalist William Howard Russell — 1857 mutiny through British colonial journalist's lens — if Indian content appears at all in MMLU, it is through a colonial/Western perspective
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Information Gaps

- Annotator demographics, IAA protocols, label-disagreement aggregation — none documented in paper.
- Whether any cross-cultural annotation comparison has been performed for MMLU labels.

Requires Expert Verification

- Indian SME re-annotation of moral_scenarios, jurisprudence, and any Indian-content items to quantify label-flip rate.
- Comparison of MMLU answer keys against UPSC/SSC official answer keys for any overlapping content.

Remediation

Gap 1: US-anchored annotator pool and US-norm scoring (especially moral_scenarios, jurisprudence, world_history); zero documented Indian SME review.

Recommendation: Conduct Indian-SME re-annotation of any MMLU items retained for STEM proxy use; for India-relevant content, rely on benchmarks with documented Indian annotation chains (ParamBench [WEB-15], MILU [WEB-13]).

ID	Evidence
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is the dimension with the strongest partial match — both MMLU and the deployment use text-based MCQ as the interaction vehicle, and the binary correct/incorrect verdict can in principle be derived from MMLU's A/B/C/D output [\[Q83\]](#). However, MMLU evaluates only single-letter classification accuracy [\[Q34, Q41\]](#); it does not evaluate any natural-language rationale, Hindi fluency, or pedagogical appropriateness. Calibration failures further reduce the value of MMLU's confidence outputs as deployment signal: GPT-3 confidence deviates from accuracy by up to 24% zero-shot [\[Q57\]](#) and 14% few-shot [\[Q96\]](#), with Elementary Mathematics RMS error of 19.4% [\[Q58\]](#). Given the deployment's negative-marking exam context where confidence calibration matters, this is a real concern. The MODERATE priority weight from elicitation reflects this partial match — score is 2 rather than 1 because the structural MCQ alignment provides some external validity.

ID	Evidence
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q41	"All values are percentages." (p.5)
Q57	"GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q58	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)
Q83	"Models are fine-tuned to predict one of four classes using the UnifiedQA MCQ questions and using our dev+val set. We test on our multitask test set." (p.12)
Q96	"While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is $r = 0.81$, compared to $r = 0.63$ in the zero-shot setting." (p.13)

Strengths

- Text-based MCQ output form is a partial match to deployment interaction vehicle [\[Q44, DATASET strength 1\]](#).
- Balanced label distribution across A/B/C/D classes prevents position-bias artifacts [\[DATASET strength 3\]](#).
- Confidence/calibration measurement framework [\[Q56, Q58\]](#) is methodologically transferable, even if the specific MMLU calibration findings are not directly applicable.

ID	Evidence
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]". For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q56	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q58	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial match — MCQ text output is appropriate, but the deployment also requires generative Hindi explanatory text which MMLU does not evaluate.

OF-2: Not applicable — deployment is text-based, not speech-based; TTS availability is not a primary requirement per elicitation.

OF-3: Literacy is not a barrier for the deployment population (graduate-level aspirants per region YAML); however, mobile-first low-bandwidth constraints apply [WEB-10] and rural shared-device usage is documented [WEB-12].

OF-4: External validity is partially harmed: a model's MMLU label-accuracy and calibration scores do not predict its capacity to produce coherent Hindi explanatory output, and calibration failures up to 24% [Q57] mean confidence signals do not transfer reliably.

ID	Evidence
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q57	"GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q96	"While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is $r = 0.81$, compared to $r = 0.63$ in the zero-shot setting." (p.13)
WEB-10	IAMAI/Kantar 2024: Bihar 43%, UP 46% internet penetration — relevant for mobile-first deployment constraints (https://www.iamai.in/sites/default/files/research/Kantar_%20IAMAI%20report_2024_.pdf)
WEB-12	https://www.business-standard.com/industry/news/india-to-exceed-900mn-internet-users-by-2025-125011600669_1.html
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)

Information Gaps

- How well MMLU's calibration framework translates to a Hindi-output context is not empirically tested.

Requires Expert Verification

- Mobile-device performance of any MMLU-evaluated model on the deployment hardware profile (mid-range Android, intermittent connectivity).

Remediation

Gap 1: Calibration failures (up to 24% off [Q57]) reduce reliability of confidence signals in a negative-marking exam context.

Recommendation: Add explicit calibration measurement on Hindi-medium UPSC/SSC items (e.g., RMS calibration error per subject) using the methodology from MMLU [Q58] but applied to Indian-curriculum benchmarks; require model abstention support for negative-marking scenarios.

ID	Evidence
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Q57	"GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q58	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%." (p.7)